(Please write your Enrollment Number)

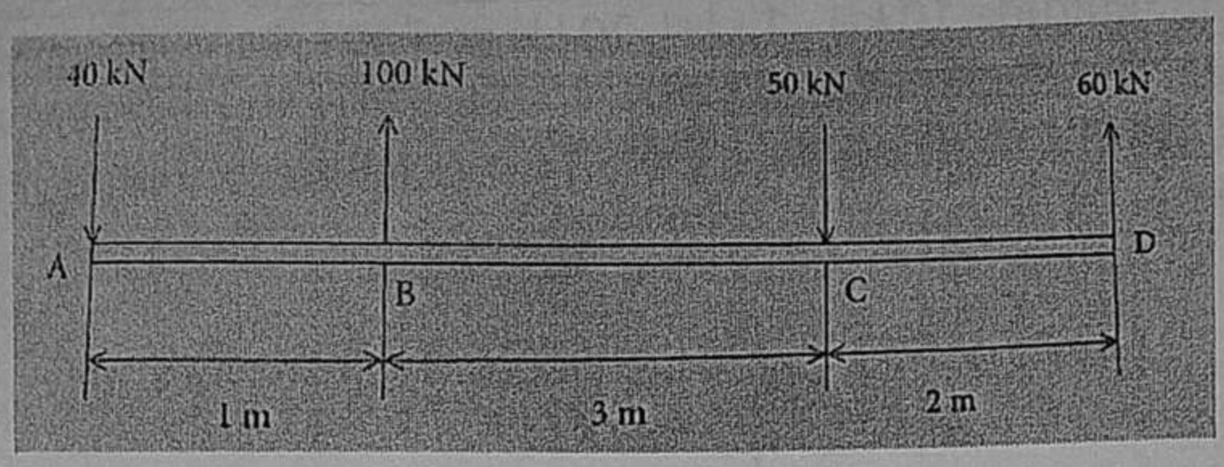
End-Term Examination (Regular and Reappear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.TECH IT/ MAE/ DMAM Semester:1
(December, 2023)

Subject Code: BMA 110	Subject: Engineering Mechanics

Time :3 Hours — Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

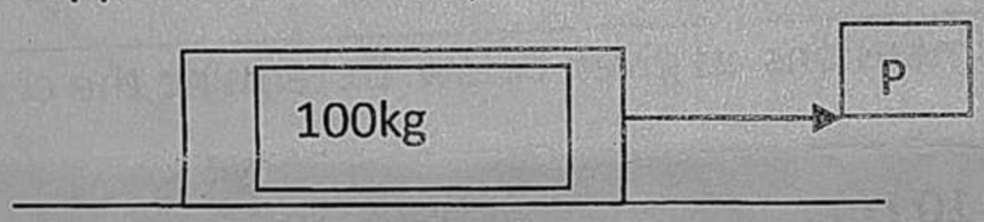
Q1		(2.5*8 =20)	
	(a) A Parallel Coplanar force system is acting on a rigid bar as shown in Fig 1		



Reduce the force system into
Single force system

(b) State the Pappus-Guldin Theorems Pappus theorem and write its expressions

(c) A wooden block rests on a horizontal surface, as shown in the figure below. If the coefficient of friction for limiting and Dynamic friction between the Block and horizontal plane are 0.4 and 0.26, respectively, Determine the status of motion if 200N force is applied Horizontally.



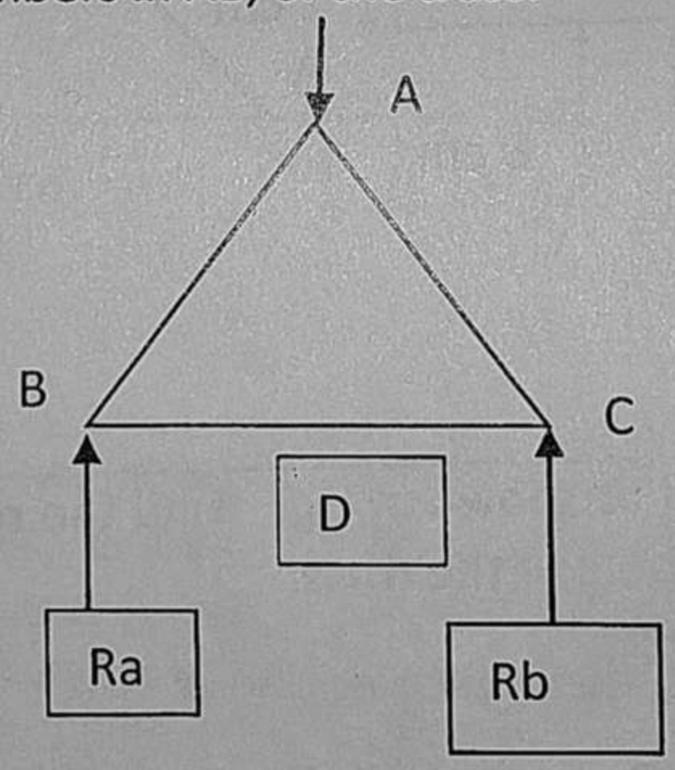
(d) Find the forces in members in AB, of the truss.

Angle ABC= 60°

Angle ACB=30°

Force at A = 30 KN

Distance AB = 2.5m



(e) What do you understand by the term "Instantaneous Centre of Rotation "

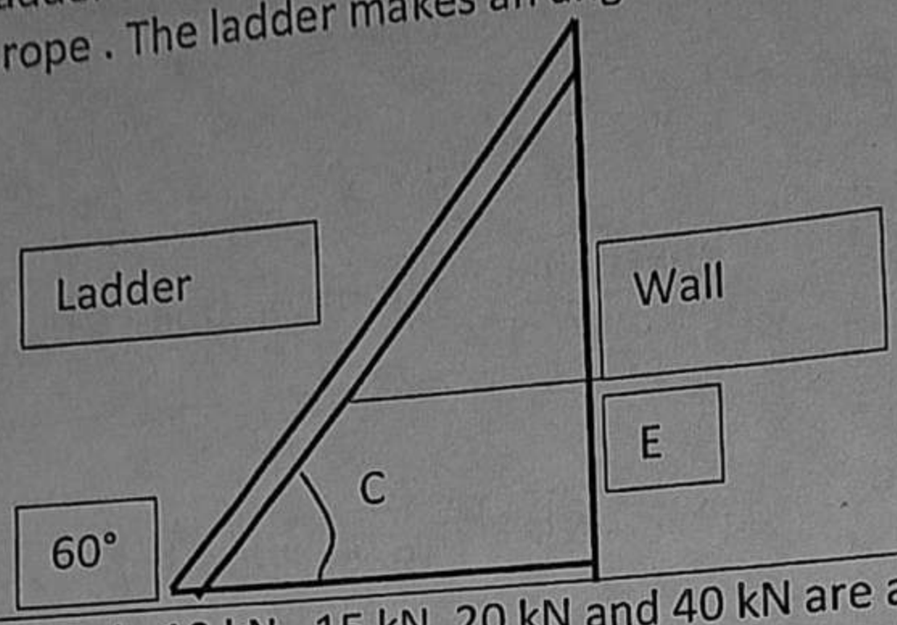
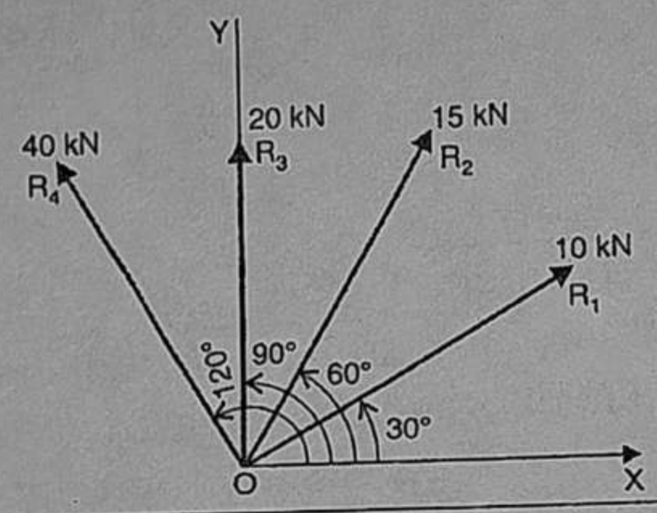
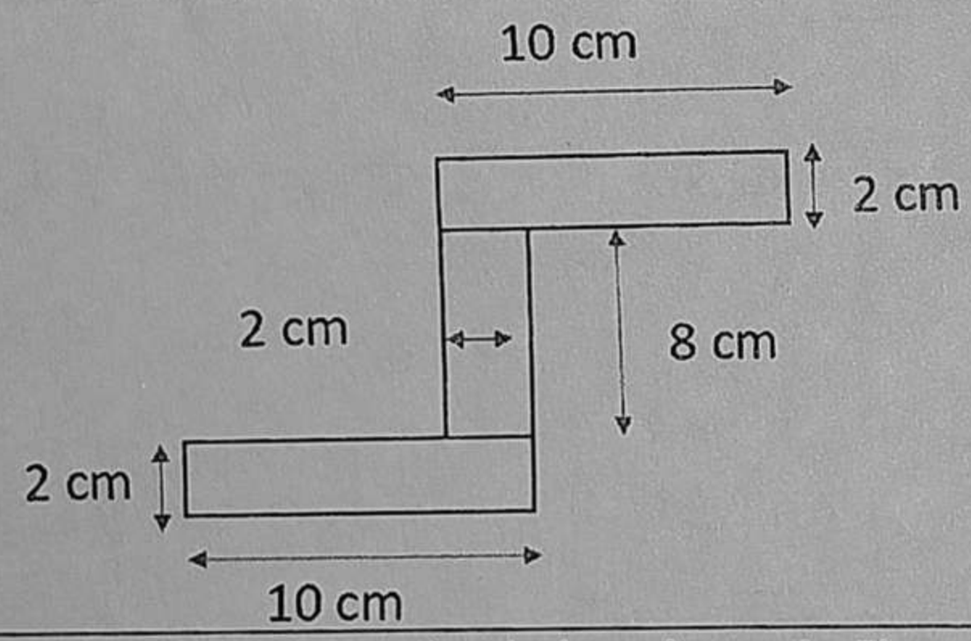
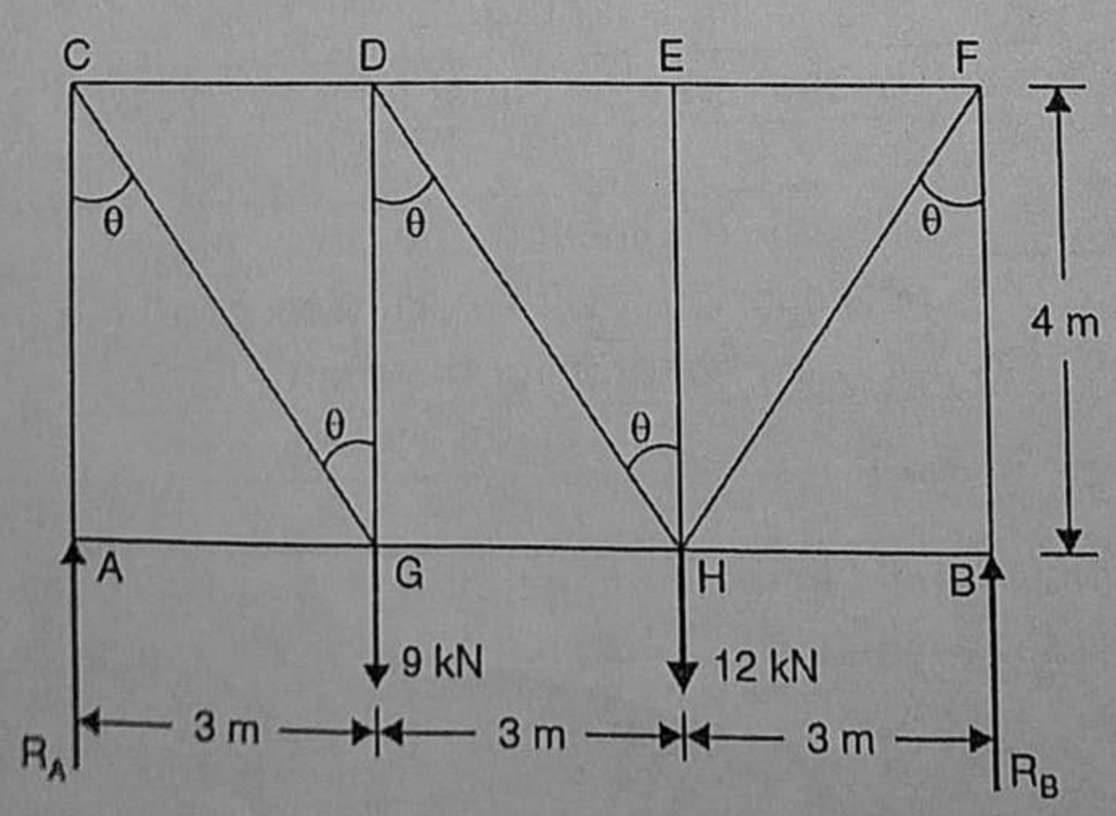
(f) State D' Alembert principle

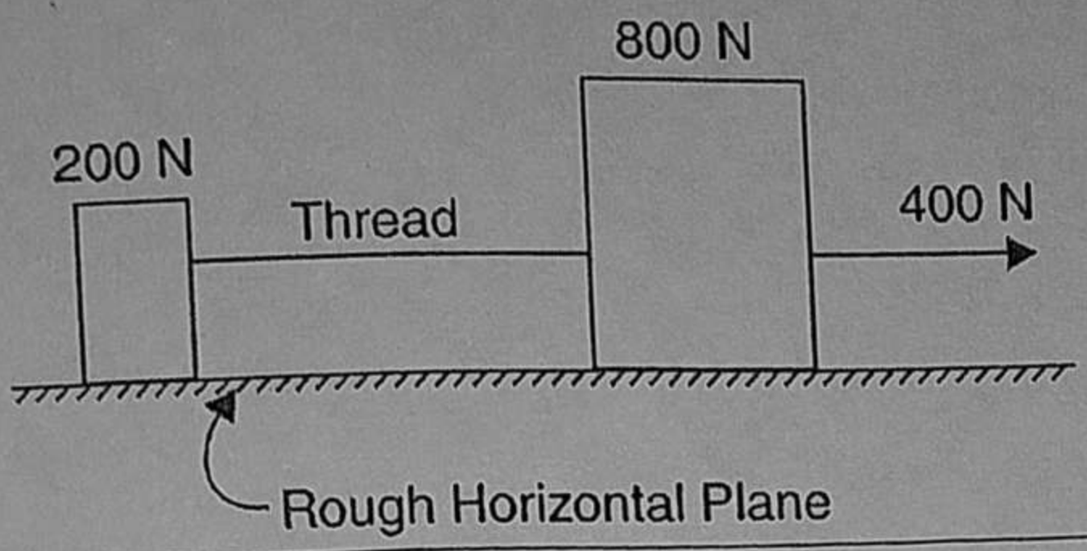
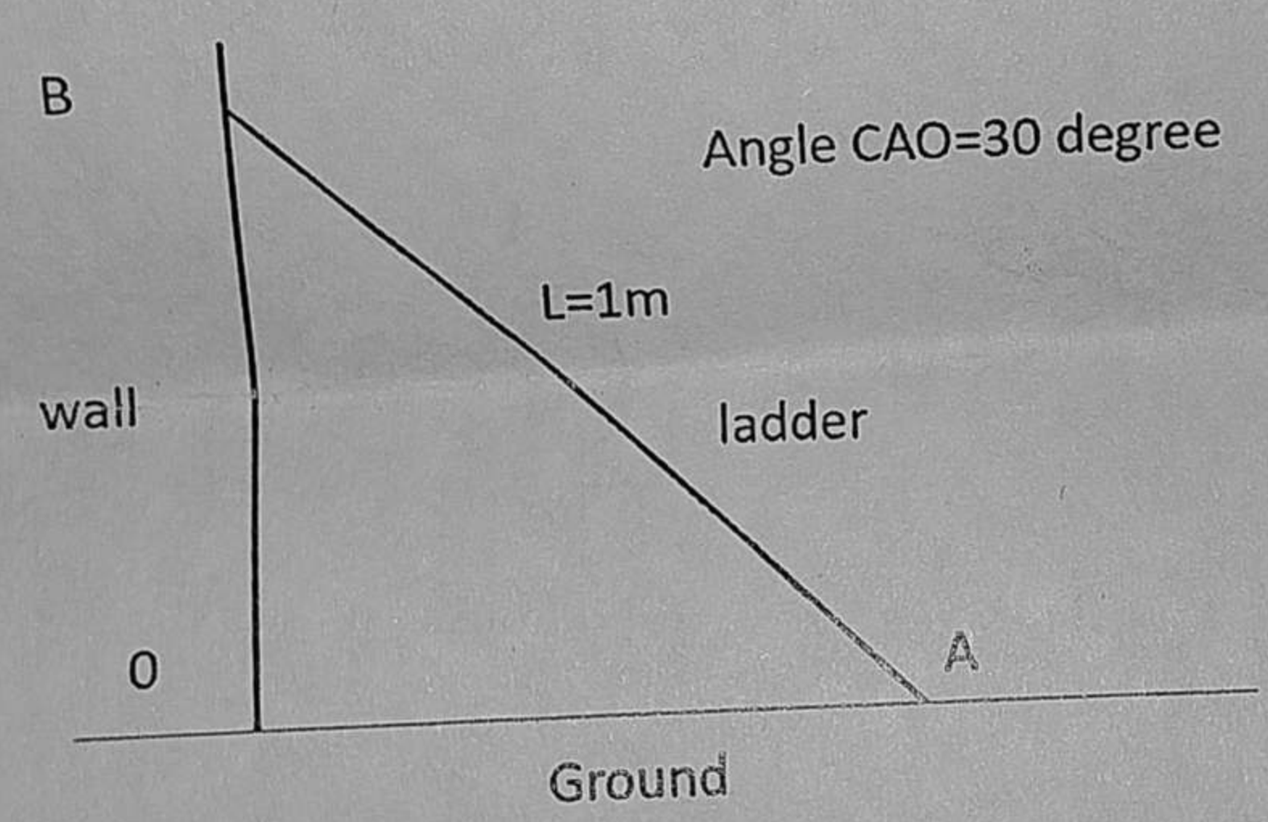
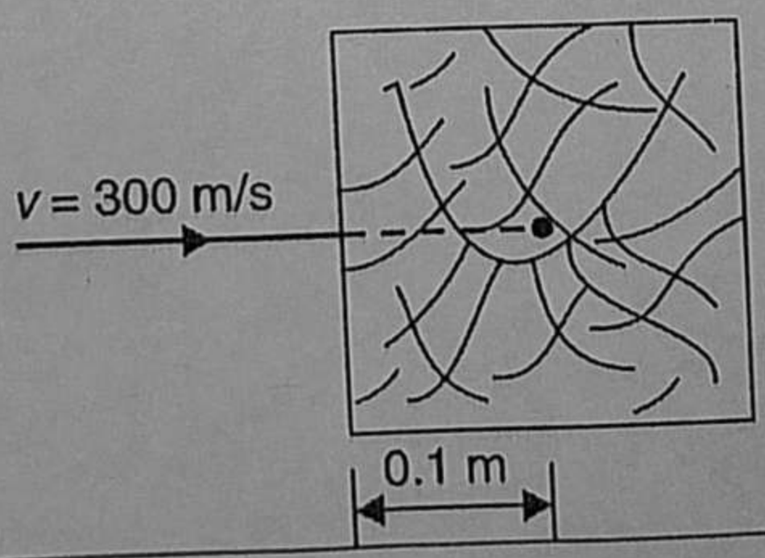
(g) Explain the work energy principle for linear motion

h) A tangential force of 1800 N acts on a shaft with a diameter of 10 mm. Find the work done by the force for one revolution of the shaft.

UNIT-I

Q2	A ladder AB of weight 1000N is held in impending motion towards the right by the rope tied to the wall as shown in figure . The coefficient of Friction between	(10)	

	the floor and the ladder is 0.3 and between the wall and ladder is 0.40 . calcula the tension in the rope . The ladder makes an angle of 60° with the ground.  		
Q3	Four forces of magnitude 10 kN , 15 kN, 20 kN and 40 kN are acting ata point O as shown in Fig.. The angles made by 10 kN, 15 kN, 20 kN and 40 kN withX-axis are 30°, 60°, 90° and 120° respectively. Find the magnitude and direction of the resultant force  	(10)	
	UNIT-II		
Q4	*Fig. shows a Z section of dimensions as given below. Determine the centroid of this Z section.*  	(10)	
Q5	A truss of span 9 m is loaded as shown in Fig. . Find the reactions and forces in the AC, AG, CD, and CG members of the truss.  	(10)	
	UNIT-III		

Q6	Two weights 800 N and 200 N are connected by a thread and move along a rough horizontal plane under the action of a force 400 N applied to the first weight of 800.N as shown in Fig. The co-efficient of friction between the sliding surfaces of the weights and the plane is 0.3. Determine the acceleration of the weights and the tension in the Thread. 	(10)	
Q7	A wheel, rotating about a fixed axis at 20 r.p.m., is uniformly accelerated for 70 seconds, during which time it makes 50 revolutions. Find : (i) Angular velocity at the end of this interval, and (ii) the Time required for the speed to reach 100 revolutions per minute	(10)	
	UNIT-IV		
Q8	An AB Bar of length 1 m has its end A and B constrained to move horizontally and vertically as shown in Fig . The end A moves with a constant velocity of 5 m/s horizontally. Find the angular velocity of the bar Ab b) velocity of end B and the velocity of midpoint C of the ladder at the instant when te bar makes an angle of 30 deg with the horizontal. 	(10)	
Q9	A bullet of mass 162 gm and moving with a velocity of 300 m/s is fired into a block of wood and it penetrates to a epth of 10 cm. If the bullet moving with the same velocity, were fired into a similar piece of wood 5 cm thick, with what velocity would it emerge ? Find also the force of resistance, assuming it to be uniform. 	(10)	Co5